

Amazon Sales Performance Analysis using SQL & Power BI

Project Overview

This project focuses on analysing Amazon sales data from February 2025 to April 2025 using PostgreSQL, Power BI, Power Query Editor, and DAX measures. The main objective of this project is to identify valuable business insights related to sales performance, customer behaviour, product categories, payment methods, and order status.

The project demonstrates practical skills in:

- SQL data analysis
- Data cleaning and transformation
- Data visualization
- Dashboard development
- KPI creation using DAX
- Business insight generation

Tools & Technologies Used

Tool	Purpose	
PostgreSQL	Data storage and SQL analysis	
Power Query Editor	Data cleaning and transformation	
Power BI	Dashboard creation and visualization	
DAX	KPI calculations and business metrics	
Kaggle	Source Dataset	

Dataset Information

The dataset contains Amazon e-commerce sales transactions from February 2025 to April 2025.

order_id ▾	Date ▾	product ▾	category ▾	price ▾	quantity ▾	total_sales ▾	customer_name ▾	customer_location ▾	payment_method ▾	status ▾
ORD0002	20 March 2025	Headphones	Electronics	100	4	400	Emily Johnson	San Francisco	Debit Card	Pending
ORD0005	10 March 2025	Smartwatch	Electronics	150	3	450	Emma Clark	New York	Debit Card	Pending
ORD0007	18 March 2025	Smartwatch	Electronics	150	4	600	Emma Clark	Houston	PayPal	Completed
ORD0008	02 March 2025	Smartphone	Electronics	500	1	500	Sophia Miller	Miami	PayPal	Completed
ORD0010	ORD0008 2025	Smartphone	Electronics	500	1	500	Emily Johnson	San Francisco	Credit Card	Cancelled
ORD0013	01 March 2025	Laptop	Electronics	800	2	1600	Daniel Harris	San Francisco	Gift Card	Pending
ORD0015	20 February 2025	Smartwatch	Electronics	150	4	600	John Doe	Seattle	Credit Card	Completed
ORD0018	10 February 2025	Smartphone	Electronics	500	2	1000	Michael Brown	Los Angeles	Amazon Pay	Completed
ORD0020	07 March 2025	Headphones	Electronics	100	4	400	Olivia Wilson	Seattle	Debit Card	Pending
ORD0021	05 February 2025	Headphones	Electronics	100	3	300	Chris White	Miami	Debit Card	Cancelled
ORD0028	03 February 2025	Headphones	Electronics	100	1	100	Jane Smith	Chicago	Amazon Pay	Completed
ORD0029	12 February 2025	Smartphone	Electronics	500	1	500	Sophia Miller	Denver	Credit Card	Cancelled
ORD0031	24 March 2025	Smartphone	Electronics	500	1	500	John Doe	Houston	Gift Card	Pending
ORD0032	10 March 2025	Smartphone	Electronics	500	4	2000	Michael Brown	Seattle	PayPal	Pending
ORD0035	02 April 2025	Laptop	Electronics	800	3	2400	Emma Clark	Denver	Amazon Pay	Completed
ORD0037	16 February 2025	Headphones	Electronics	100	3	300	Michael Brown	New York	Debit Card	Cancelled
ORD0038	26 March 2025	Laptop	Electronics	800	3	2400	Olivia Wilson	Chicago	Amazon Pay	Completed
ORD0039	02 March 2025	Laptop	Electronics	800	2	1600	Olivia Wilson	San Francisco	PayPal	Completed
ORD0040	26 March 2025	Smartwatch	Electronics	150	1	150	Emily Johnson	Seattle	Gift Card	Pending
ORD0042	17 March 2025	Headphones	Electronics	100	3	300	Jane Smith	Chicago	Amazon Pay	Cancelled
ORD0044	24 March 2025	Smartwatch	Electronics	150	1	150	Chris White	Houston	Debit Card	Pending
ORD0049	18 February 2025	Smartphone	Electronics	500	4	2000	Emma Clark	Chicago	Debit Card	Completed
ORD0052	24 February 2025	Laptop	Electronics	800	3	2400	John Doe	San Francisco	Credit Card	Pending
ORD0054	15 March 2025	Smartwatch	Electronics	150	3	450	David Lee	Boston	Gift Card	Pending
ORD0056	19 March 2025	Smartwatch	Electronics	150	2	300	Emma Clark	Dallas	Credit Card	Completed
ORD0057	15 March 2025	Smartphone	Electronics	500	1	500	Jane Smith	Los Angeles	Debit Card	Cancelled
ORD0058	13 March 2025	Smartphone	Electronics	500	1	500	Jane Smith	Chicago	PayPal	Cancelled
ORD0059	01 April 2025	Smartwatch	Electronics	150	2	300	Daniel Harris	Dallas	Credit Card	Cancelled
ORD0062	10 February 2025	Laptop	Electronics	800	5	4000	Olivia Wilson	San Francisco	PayPal	Completed
ORD0063	30 March 2025	Smartphone	Electronics	500	5	2500	Emma Clark	Miami	Gift Card	Completed
ORD0066	14 March 2025	Smartwatch	Electronics	150	2	300	Michael Brown	Denver	PayPal	Pending
ORD0067	31 March 2025	Headphones	Electronics	100	3	300	Chris White	New York	Debit Card	Pending

Project Workflow

Step 1: Data Import into PostgreSQL

The dataset was imported into PostgreSQL database for performing SQL analysis.

A SQL table was created to store the sales data.

Example SQL query:

Query	Query History
1	CREATE TABLE amazon_sales (2 order_id VARCHAR(20), 3 order_date VARCHAR(20), 4 product VARCHAR(100), 5 category VARCHAR(50), 6 price NUMERIC, 7 quantity INT, 8 total_sales NUMERIC, 9 customer_name VARCHAR(100), 10 customer_location VARCHAR(100), 11 payment_method VARCHAR(50), 12 status VARCHAR(20) 13); 14 15 SELECT * FROM amazon_sales 16 LIMIT 10;

Step 2: SQL Data Analysis

Example SQL Queries

Top Revenue

```
-- What is the total revenue generated?  
  
SELECT SUM (total_sales) AS total_revenue  
FROM amazon_sales;
```

Output	Messages	Notifications
total_revenue numeric 243845		

Revenue by Category

27

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-- Which product category generates the highest sales?

SELECT category,

SUM(total_sales) AS revenue

FROM amazon_sales

GROUP BY category

ORDER BY revenue DESC;

Data Output

Messages

Notifications

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SQL

	category character varying (50) 🔒	revenue numeric 🔒
1	Electronics	129950
2	Home Appliances	105000
3	Footwear	4320
4	Clothing	3540
5	Books	1035

Top revenue by city

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-- Which city generates the highest revenue?

SELECT customer_location,

SUM(total_sales) AS revenue

FROM amazon_sales

GROUP BY customer_location

ORDER BY revenue DESC

LIMIT 5;

Data Output

Messages

Notifications

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SQL

	customer_location character varying (100) 🔒	revenue numeric 🔒
	Miami	31700
	Denver	29785
	Houston	28390
	Dallas	27145
	Seattle	26890

Top selling product in amazon

```
35
36 -- Which products are sold the most?
37
38 SELECT product,
39        SUM (quantity) AS total_quantity_sold
40 FROM amazon_sales
41 GROUP BY product
42 ORDER BY total_quantity_sold DESC;
43
44
45
46
```

Data Output Messages Notifications

	product character varying (100) 🔒	total_quantity_sold bigint 🔒
1	Smartwatch	105
2	Smartphone	97
3	Headphones	73
4	Laptop	73
5	Running Shoes	72
6	Book	69
7	Refrigerator	65
8	Jeans	62
9	T-Shirt	53
10	Washing Machine	45

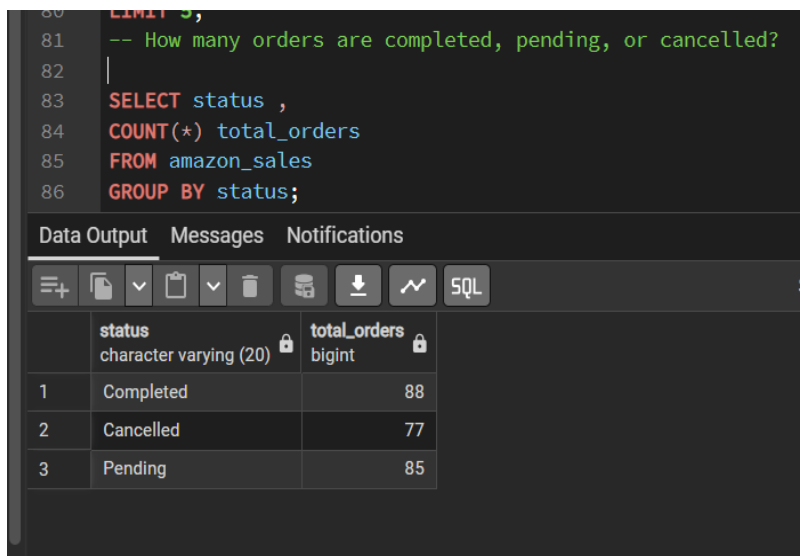
Top payment method

```
60
61 -- Which payment method is used the most?
62
63 SELECT payment_method,
64        COUNT(*) AS total_orders
65 FROM amazon_sales
66 GROUP BY payment_method
67 ORDER BY total_orders DESC;
68
```

Data Output Messages Notifications

	payment_method character varying (50) 🔒	total_orders bigint 🔒
1	PayPal	60
2	Credit Card	54
3	Debit Card	53
4	Gift Card	42
5	Amazon Pay	41

Order status analysis



The screenshot shows a SQL query editor with a dark theme. The query is as follows:

```
80 LIMIT 5;  
81 -- How many orders are completed, pending, or cancelled?  
82 |  
83 SELECT status ,  
84 COUNT(*) total_orders  
85 FROM amazon_sales  
86 GROUP BY status;
```

Below the query editor, there is a toolbar with icons for saving, undo, redo, and other functions. The results are displayed in a table with the following data:

	status character varying (20)	total_orders bigint
1	Completed	88
2	Cancelled	77
3	Pending	85

Step 3: Data Cleaning using Power Query Editor

The dataset was cleaned and transformed using Power Query Editor in Power BI.

Cleaning Steps Performed

- Checked missing values
- Corrected data types
- Converted date column into proper date format
- Removed unnecessary spaces
- Validated numerical columns
- Prepared dataset for visualization

Step 4: DAX Measures Creation

DAX measures were created to calculate important KPIs and business metrics.

AVG order value

Structure	Formatting	Properties
✓	1 AVG Order value = AVERAGE('public amazon_sales'[total_sales])	
id	Date	product category price quantity total sales

Count of order cancelled

```
1 Cancelled Orders = CALCULATE(COUNT('public amazon_sales'[order_id]),'public amazon_sales'[status] = "Cancelled")
```

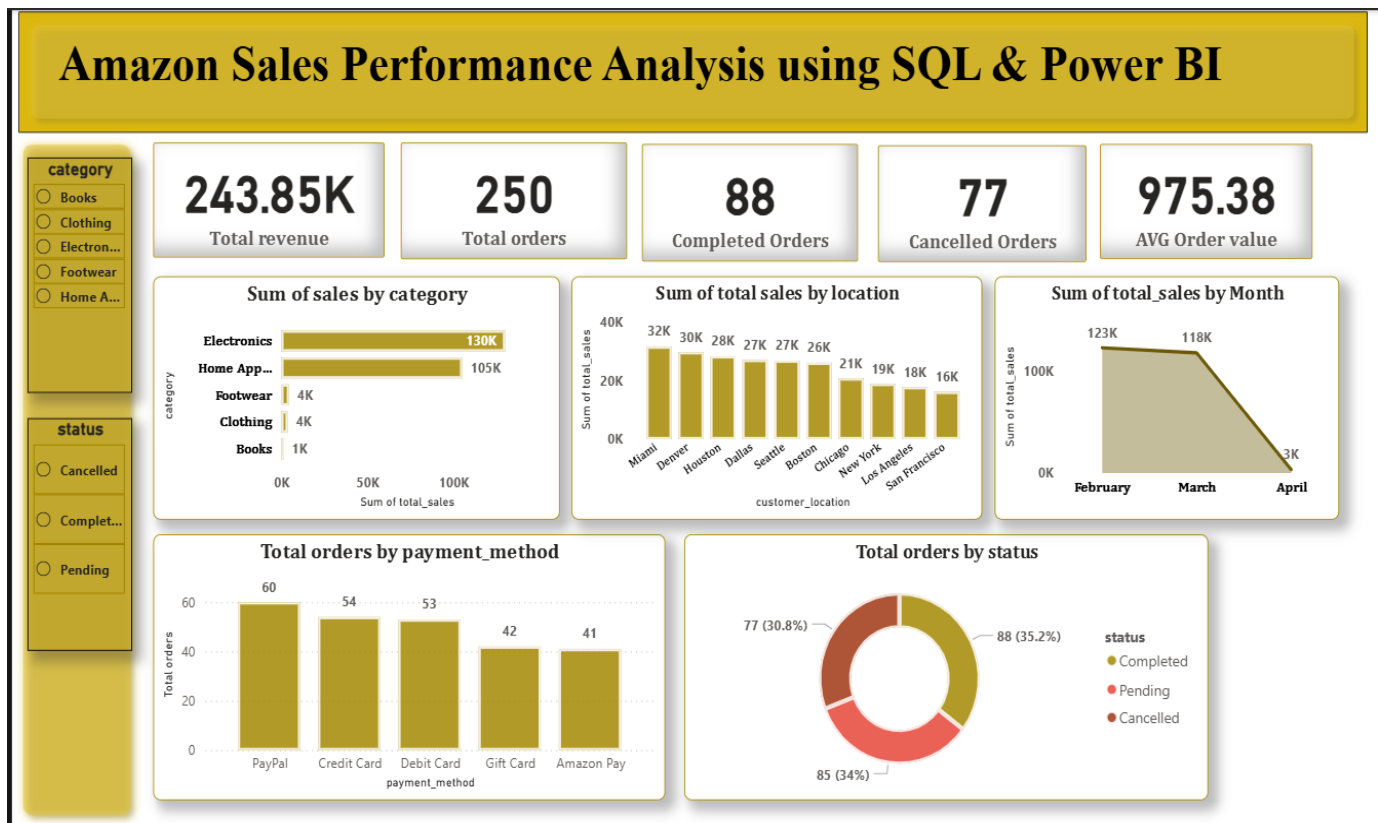
Completed orders

```
1 Completed Orders = CALCULATE(COUNT('public amazon_sales'[order_id]),'public amazon_sales'[status] = "Completed")
```

Count of total orders

Structure	Formatting	Properties
✓	1 Total orders = COUNT('public amazon_sales'[order_id])	

Dashboard Development in Power BI



An interactive dashboard was developed in Power BI to visualize sales performance and business insights.

KPI Cards

- Total Revenue
- Total Orders
- Completed Orders
- Cancelled Orders
- Average Order Value

Visualizations Used

- Revenue by Category
- Revenue by Customer Location
- Monthly Sales Trend
- Payment Method Analysis

- Order Status Analysis

Filters/Slicers Added

- Category
- Order Status

Key Business Insights

1. Electronics Generated the Highest Revenue

The Electronics category contributed the largest share of total sales revenue, indicating strong customer demand for electronic products.

2. Home Appliances Showed Strong Sales Performance

Home Appliances was the second-highest revenue generating category.

3. PayPal Was the Most Preferred Payment Method

Customers frequently used PayPal for transactions compared to other payment methods.

4. Completed Orders Dominated Transactions

Most customer orders were successfully completed, showing positive operational performance.

5. Certain Locations Generated Higher Revenue

Cities such as Miami, Denver, and Houston contributed significantly higher sales revenue.

6. Revenue Declined in April 2025

Monthly sales analysis showed a significant drop in revenue during April 2025

Conclusion

This project successfully analysed Amazon sales data using PostgreSQL and Power BI. SQL queries were used for data analysis, Power Query Editor was used for cleaning and transformation, and DAX measures were used to create meaningful KPIs.

The final Power BI dashboard provided valuable business insights regarding:

- Revenue performance
- Product category trends
- Customer purchasing behaviour
- Payment method preferences
- Order completion analysis

This project helped improve practical skills in SQL, Power BI, data cleaning, dashboard development, and business analysis.

Skills Demonstrated

- SQL Query Writing
- PostgreSQL Database Management
- Data Cleaning
- Power Query Transformation
- DAX Calculations
- Data Visualization
- Dashboard Design
- Business Insight Generation
- Analytical Thinking

Future Improvements

Future improvements for this project may include:

- Adding forecasting analysis
- Customer segmentation analysis
- Profit margin analysis
- Advanced DAX calculations
- Interactive drill-through reports
- Real-time database integration

Project Title

Amazon Sales Performance Analysis using SQL & Power BI

created a complete professional project report for your Amazon Sales Analysis dashboard including:

- Project overview
- Tools used
- SQL analysis
- Data cleaning process
- DAX measures
- Dashboard explanation
- Business insights
- Conclusion
- Skills demonstrated

This is suitable for:

- Portfolio projects
- LinkedIn showcase
- Resume projects section
- Internship/job applications
- GitHub documentation